



COMP5310 – Project Stage 2

Lab08_Group_09

SID: 550891659, Unikey: hjos0419

SID: 550832272, Unikey: tgar1036

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DATASET C: HOTEL BOOKINGS

1 Topic Selection and Problem Definition

1.1 Dataset Selection

- For Stage 2, we selected the Hotel Booking Dataset for further analysis and modelling. The dataset contains hotel reservation records from 2015–2017, including booking details, customer characteristics, pricing information, and cancellation outcomes.
- It was chosen because it presents a clear and highly relevant business problem, predicting hotel booking cancellations, while offering the combination of data quality, interpretability, business applicability, and suitability for analytical modelling compared to the other two datasets.
- Stage 1 exploratory analysis revealed several meaningful relationships between booking characteristics and cancellation behaviour. Features such as lead_time, deposit_type, adr, market_segment, and previous_cancellations demonstrated clear behavioural patterns associated with booking cancellations, indicating strong potential for predictive analysis.
- Compared to the NSW Train Occupancy dataset, the hotel dataset involved fewer operational complexities and less severe class imbalance, making it more appropriate for reliable model evaluation and interpretation. The train dataset also relied heavily on time-dependent operational behaviour, increasing modelling complexity.
- Compared to the Airline Delay dataset, the hotel dataset presented a lower risk of target leakage and included more stable booking-time features that are available before the outcome occurs. In contrast, several airline delay variables were extremely strongly correlated with the target variable, which could artificially inflate predictive performance without improving practical decision-making.
- Additionally, it is a mixture of numerical and categorical variables enables meaningful comparison between classification and clustering approaches. Overall, the Hotel Booking Dataset was selected because it provides a realistic business problem, strong analytical potential, and a suitable foundation for applying multiple data-driven modelling techniques.

1.2 Summary of the Research Problem from Stage 1

The research problem identified in Stage 1 centres on the **financial and operational impact of booking cancellations** in the hospitality industry. When a guest cancels, the hotel suffers "revenue leakage" from unoccupied rooms and incurs costs associated with over-staffing and wasted inventory.

The problem was defined as a **Binary Classification task**: Using guest data available at the moment of booking to predict the target variable `is_canceled` (1 if cancelled, 0 if fulfilled). The objective is to move the hotel from a reactive state to a proactive one, allowing management to anticipate cancellations before they happen.

1.3 Refined Research Problem

For Stage 2, We plan to refine the problem to focus on **Multi-Dimensional Risk Profiling**.

Updated Research Question:

"How can the hotel accurately predict the probability of cancellation, and what distinct 'risk personas' exist among guests that contribute to revenue instability?"

To answer this, we will implement **three specific models**:

1. **Supervised Model A (Random Forest Classifier)**: To achieve high predictive accuracy and extract "Feature Importance" to identify the strongest predictors of cancellation.
2. **Supervised Model B (Logistic Regression)**: To establish a clear, interpretable baseline and understand the direct relationship between features like `lead_time` or `adr` and the likelihood of cancellation.
3. **Unsupervised Model C (K-Means Clustering)**: To segment cancelled bookings into groups to see if specific guest behaviours cluster together.

1.3 Justification and Motivation

A. Motivation for the Dataset Choice

- The Hotel Booking dataset was chosen because it enables the development of practical and actionable business solutions within the hospitality industry. Booking cancellations directly impact hotel revenue, occupancy forecasting, staffing allocation, and customer relationship management, making cancellation prediction an important operational challenge.
- A key advantage of this dataset is that most variables are available at the time the reservation is made. This allows predictive models to identify high-risk bookings before cancellations occur, enabling proactive business strategies such as dynamic deposit policies, targeted retention offers, and improved overbooking management.
- The dataset is also well suited for applying multiple analytical methods with different objectives. Logistic Regression provides interpretable insights into how individual variables influence cancellation probability, Random Forest improves predictive accuracy and identifies the most influential features, while K-Means clustering supports behavioural segmentation of customer booking patterns.
- Overall, the dataset supports both predictive performance and business interpretability, making it highly suitable for building a comprehensive cancellation risk analysis framework.

B. Motivation for the Problem Definition

1. **From Prediction to Persona**: Simply knowing a booking *might* cancel is only half the battle. By adding the unsupervised clustering model, we identify **who** is cancelling. This allows the marketing team to tailor different retention strategies for different groups.

2. **Addressing the "Strongest Predictors":** By using Random Forest, we can pinpoint exactly which "levers" the hotel needs to pull. For example, if deposit_type is the #1 predictor, the hotel has a clear business case to mandate non-refundable deposits for high-risk lead times.
3. **Operational Efficiency:** This definition directly addresses **Resource Allocation**. With a high precision supervised model, the hotel can confidently "overbook" specific room types, knowing exactly which ones are likely to become vacant, thus maximizing the occupancy rate and Average Daily Rate (ADR).

1.4 Conclusion

By limiting the analysis to these three specific models (two supervised for accuracy and interpretability, and one unsupervised for behavioural insight), Stage 2 will provide a comprehensive "Cancellation Prevention Framework" that is grounded in the deep patterns of the provided dataset.

2 Data Description

2.1 Dataset Overview

The hotel bookings dataset covers 119,987 reservations from two Portuguese hotels (a City Hotel and a Resort Hotel) spanning July 2015 to August 2017. After the full Stage 1 cleaning pipeline removing 31,328 exact duplicates, eliminating zero-guest invalid entries, correcting negative ADR values, imputing missing values, and normalising string inconsistencies, the working dataset contains 87,678 rows and 31 columns including engineered features.

Property	Value
Final rows (after cleaning)	87,678
Final columns (after cleaning)	31
Date range	2015 – 2017
Target: is_canceled = 0 (Not Cancelled)	63,381 (72.3%)
Target: is_canceled = 1 (Cancelled)	24,297 (27.7%)
Missing values (post-cleaning)	None
Duplicate rows removed	31,328

The class distribution (72.3% not cancelled, 27.7% cancelled) is mildly imbalanced but does not require resampling techniques. We address it by applying class_weight='balanced' in supervised models and using stratified train/test splitting, and by reporting F1 score, Precision, and Recall alongside Accuracy.

2.2 Key challenges and implication

Stage 1 surfaced several data quality and structural challenges, each of which has been handled prior to modelling:

Challenge	Detail	Implication for Modelling
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Data leakage	reservation_status and reservation_status_date encode the outcome directly	Both columns dropped — they only exist after cancellation occurs and would invalidate model evaluation
Class imbalance	72.3% not cancelled vs 27.7% cancelled	Accuracy alone is misleading; class_weight='balanced' applied; F1, Precision, Recall reported
High-cardinality feature	country has 178 unique values	Dropped — ordinal/one-hot encoding of 178 classes adds noise with negligible predictive gain at booking time
Agent ID as numeric	agent stored as integer ID, not a true numeric	Converted to binary flag is_direct_booking (1 = no agent, 0 = via agent) to remove spurious ordinal meaning
Date format inconsistency	reservation_status_date had 955 rows with non-ISO formats	Resolved during Stage 1; column subsequently dropped due to leakage
Redundant stay features	stays_in_weekend_nights and stays_in_week_nights captured separately	Consolidated into engineered feature total_stay_nights to reduce multicollinearity for Logistic Regression

2.3 Intended Variables

A. The Target Variable

Variable Name: is_canceled

Type: Binary (1 = Cancelled/No-show; 0 = Check-out/Fulfilled).

Justification: This variable is the direct answer to our research question. From a business perspective, predicting this outcome allows for revenue recovery through strategic overbooking and dynamic cancellation policies.

The dataset has significant class imbalance in the target variable is_canceled:

- **Not Cancelled (0):** 72.29%
- **Cancelled (1):** 27.71%

This means that only about 27.71% of the bookings were cancelled.

B Feature Variables and Justification

Feature Variables: The following features have been selected to build the predictive models. This selection is based on their potential influence on booking cancellations as observed in common hotel booking datasets and preliminary data understanding.

The selected features are categorized into three "Risk Pillars" to ensure the model captures different dimensions of guest behaviour.

Pillar	Variables	Justification based on Stage 1 Findings
Commitment & Intent	lead_time, total_of_special_requests, required_car_parking_spaces, booking_changes	Lead Time was found to have a high variance; bookings made far in advance are psychologically more "disposable." Conversely, guests who request parking or make special requests demonstrate higher "investment" in the stay, significantly lowering cancellation probability.
Financial Stake	adr (Average Daily Rate), deposit_type, customer_type	Deposit Type is a critical predictor; "Non Refund" bookings act as a strong deterrent to cancellation. ADR allows the model to calculate the "cost of walking away" and higher-priced rooms may be canceled if the guest finds a last-minute deal elsewhere.
Trust & History	previous_cancellations, is_repeated_guest, previous_bookings_not_canceled	These features capture the guest's historical reliability. A guest with a high previous_cancellations count demonstrates a behavioral pattern that is likely to repeat, whereas is_repeated_guest signals brand loyalty and higher fulfillment rates.
Contextual	hotel, market_segment, distribution_channel, meal	These define the "Type" of booking. Stage 1 identified that "Online TA" (Travel Agents) often have higher cancellation rates compared to "Direct" bookings due to the ease of clicking "cancel" on third-party apps.

Derived and Engineered Features

To maximize the accuracy of the supervised models and the logic of the unsupervised clustering, we will use the five features engineered in Stage 1:

- total_stay_nights: (Numerical) Sum of weekend and week nights.
 - Justification:* Reduces dimensionality while capturing the "total financial commitment" and stay complexity.
- total_guests: (Numerical) Sum of adults, children, and babies.

- *Justification:* Solo travelers and large groups have different risk profiles; this metric captures the "operational load" of the booking in one value.
- 3. `is_family`: (Binary) Flag for presence of children/babies.
 - *Justification:* Families are a specific leisure segment. While they plan more, they are sensitive to external factors (health/emergencies), creating a distinct cluster of behavior.
- 4. `stay_category`: (Categorical) Weekend only, Weekday only, or Both.
 - *Justification:* Acts as a proxy for visit purpose. "Weekday only" stays signal business travel (shorter lead times, high frequency), while "Both" signals leisure travel (longer lead times, lower frequency).
- 5. `room_type_changed`: (Binary) Reserved vs. Assigned room discrepancy.
 - *Justification:* Captures "Operational Intervention." If a hotel changes a room type before arrival, it alters guest expectations, potentially impacting the decision to fulfill the booking.

Justification for Feature Selection: These features cover various aspects of a booking, including booking behavior, financial commitment, customer segment and specific requirements. They are all directly or indirectly related to a customer's likelihood of canceling a booking.

3. Modelling and Analysis

We will apply three distinct analytical techniques to the dataset to predict and understand cancellations:

3.1 Logistic Regression

A. Method Description

How it works: Logistic Regression predicts the probability of a binary class (e.g., Canceled vs. Not Canceled) by fitting data to a logistic (sigmoid) curve. The output is a value between 0 and 1, representing the probability of the positive class.

Key Assumptions:

- **Linearity in log-odds:** The relationship between each feature and the log-odds of cancellation is assumed to be linear.
- **Independence of observations:** Each booking record is treated as an independent observation. This is approximately satisfied since the dataset contains bookings from different guests across different time periods.
- **No severe multicollinearity:** Correlated features destabilise coefficient estimates. This was addressed by dropping `stays_in_weekend_nights` and `stays_in_week_nights` in favour of the engineered `total_stay_nights`, and by applying L1/L2 regularisation.

Strengths:

- Full interpretability: standardised coefficients reveal both the direction (positive = increases cancellation risk) and relative magnitude of each feature's effect, providing actionable insight for hotel managers.
- Naturally outputs calibrated probabilities, directly answering the research question's requirement to predict the probability of cancellation.
- Extremely stable: very low variance across cross-validation folds (std ≈ 0.003), as expected from a parametric linear model.

Limitations: Cannot easily capture complex, non-linear relationships; performance degrades if the decision boundary is highly non-linear.

B. Justification of Use

Type of problem: Binary classification. Logistic Regression is the canonical interpretable baseline for binary classification and directly outputs probabilities, the exact quantity named in the research question.

Nature of features: After one-hot encoding, all features become numerical. StandardScaler ensures that the gradient-based optimiser (liblinear) converges efficiently and that coefficients are directly comparable in magnitude.

Critically, LR provides the directional insight. While RF tells us that lead_time is the most important feature, LR tells us that increasing lead_time increases cancellation probability, a distinction that is essential for translating model output into hotel policy.

C. Application Strategy

Feature Selection: All features after one-hot encoding (9 categorical $\rightarrow$ 40 dummy variables, plus 21 numerical features = ~ 61 total features after drop_first=True).

Preprocessing: One-hot encoding with drop_first=True to avoid the dummy variable. StandardScaler fitted on training data only and applied to both train and test sets, preventing data leakage. Stratified 80/20 split identical to RF.

Hyperparameters:

Hyperparameter	Values Searched	Rationale
C (regularisation)	0.01, 0.1, 1, 10	Inverse regularisation strength. Lower C = stronger penalty. C=0.01 over-regularises (F1 drops sharply). C=0.1 with L1 achieves best CV F1.
penalty	l1, l2	L1 produces sparse models (irrelevant coefficients $\rightarrow$ 0); L2 shrinks all. L1 at C=0.1 slightly outperforms L2, producing a cleaner model.
solver	'liblinear' (fixed)	The only scikit-learn solver supporting both L1 and L2 penalties; efficient for this dataset size.

class_weight	'balanced' (fixed)	Consistent with RF for fair comparison; adjusts for class imbalance without resampling.
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GridSearchCV runs on the full training set (LR is fast enough), using 3-fold StratifiedKFold and F1 as the scoring metric, consistent with RF.

3.2 Random Forest

A. Method Description

How it works: An ensemble method that constructs a multitude of decision trees during training. It uses 'bagging' (Bootstrap Aggregating) to train trees on different subsets of data and features, then outputs the mode (classification) or mean (probability) of the individual trees.

Key Assumptions:

- Minimal assumptions; does not require features to be normally distributed or linearly related to the target.
- **Feature independence is NOT assumed:** RF handles correlated features by randomly subsetting them at each split, which was important here given that lead_time, total_stay_nights, and arrival date components are partially correlated.

Strengths:

- Consistently high predictive accuracy on tabular classification tasks, the ensemble reduces variance without increasing bias.
- Built-in feature importance (Mean Decrease in Impurity) provides a ranked list of the strongest predictors, directly answering the second component of the research question.
- Robust to outliers, scale differences, and missing patterns, split thresholds are based on rank ordering, not raw values, so no scaling is needed.
- Handles mixed feature types natively after ordinal encoding, without requiring one-hot expansion.

Limitations:

- Limited interpretability: an ensemble of 200 trees cannot be expressed as human-readable rules, unlike a single decision tree or Logistic Regression coefficients.
- Gini importance bias: continuous features with many unique values (e.g., lead_time, adr) tend to receive inflated importance scores relative to binary or low-cardinality features, potentially understating the role of deposit_type.
- Computationally heavier than Logistic Regression: training and hyperparameter search are slower, necessitating a subsampled GridSearch strategy.

B. Justification of Use

Type of problem: Binary classification: predicting is_canceled $\in \{0, 1\}$. Random Forest is among the most consistently high-performing algorithms for tabular binary classification and is a natural fit for this task.

Nature of features: The dataset contains a rich mix of 21 numerical features (lead_time, adr, total_stay_nights, etc.) and 9 categorical features (hotel, deposit_type, market_segment, etc.). RF handles this mix natively after ordinal encoding, without requiring the extensive preprocessing that Logistic Regression demands.

The primary justification for RF is its dual role: it simultaneously serves as the highest-accuracy classifier and the feature importance engine. Stage 1 EDA identified lead_time and deposit_type as visually strong predictors, RF allows these signals to be quantified and ranked across all 30 features at once, producing the definitive answer to which booking characteristics most strongly predict cancellation.

C. Application Strategy

Feature Selection: All features from the master cleaned dataset are included. RF's internal feature selection via random subsetting means uninformative features are effectively down-weighted without manual pre-filtering.

Preprocessing: OrdinalEncoder applied to 9 categorical columns. No scaling applied, RF is invariant to monotonic feature transformations. Stratified 80/20 train/test split preserves the 72.3/27.7 class ratio.

Hyperparameters:

Hyperparameter	Values Searched	Rationale
n_estimators	100, 200	More trees reduce variance; 200 is the practical ceiling before diminishing returns on 87k rows.
max_depth	10, 20, None	Controls tree depth and overfitting. None = fully grown; results show None causes severe overfitting (Train F1 $\approx$ 0.997, CV F1 drops).
min_samples_split	2, 5	Minimum samples required to split a node. Higher values increase regularisation and generalisation.
max_features	'sqrt' (fixed)	Standard best practice for classification RF — each split considers $\sqrt{n}$ features, decorrelating trees.
class_weight	'balanced' (fixed)	Adjusts weights inversely proportional to class frequency, compensating for the 72/28 imbalance.

GridSearchCV is run on a stratified 25,000-row subsample of the training set (3-fold StratifiedKFold, scoring='f1'). The best parameter combination is then refitted on the full 70,142-row training set before evaluation, ensuring the final model benefits from all available training data.

3.3 K-Means Clustering

A. Method Description

How it works: Partitions the dataset into K (4 in this case) pre-defined, non-overlapping clusters. It iteratively assigns each data point to the cluster with the nearest mean (centroid) and updates the centroids until convergence.

Key Assumptions: Assumes clusters are spherical and have similar variance; sensitive to feature scales (requires normalization).

Strengths: Efficient for large datasets; provides clear, distinct groupings that can be translated into business segments. Complementary to the supervised models — addresses a genuinely different analytical question (segmentation rather than prediction).

Limitations: Requires the user to choose 'K' in advance; sensitive to initial centroid placement and outliers.

B. Justification of Use

Type of problem: Unsupervised clustering — no target label is used. This is appropriate here because the sub-question ('what types of cancellers exist?') has no ground-truth answer and requires exploratory, data-driven segmentation.

Nature of features: After ordinal encoding and StandardScaler, all features are numeric and on a comparable scale. Outlier capping at the 99th percentile for `adr`, `lead_time`, and `days_in_waiting_list` prevents extreme values from distorting centroids.

The choice to apply K-Means only to cancelled bookings (rather than the full dataset): we are not trying to separate cancellations from non-cancellations (the supervised models do this), but to understand the heterogeneity within the cancelled group.

C. Application Strategy

Feature Selection: Applied exclusively to the subset of rows where `is_canceled == 1`. It uses all features available to define the dimensions of the 'risk personas'.

Preprocessing: Requires **Standard Scaling** on all input features because K-Means calculates Euclidean distance, which is highly sensitive to feature magnitude.

Hyperparameters:

Parameter	Values Evaluated	Finding
<code>n_clusters (k)</code>	2, 3, 4, 5, 6, 7, 8	Elbow bend and silhouette peak both at $k=4$. Confirmed as optimal.

init method	k-means++, random	k-means++ consistently achieves lower inertia and higher silhouette than random initialisation.
n_init	10, 20, 50	n_init=10 already produces a stable solution; n_init=20 and 50 yield identical silhouette. n_init=10 selected.

4. Experimental Setup and Results

4.1 Experimental setup

To ensure the validity and reproducibility of the predictive models and clustering analysis, the following experimental framework was established:

A. Data Preprocessing & Feature Engineering

- **Missing Value Handling:** The dataset was confirmed to have no missing values, requiring no imputation.
- **Feature Engineering:** Several domain-specific features were utilized to improve model sensitivity including total_stay_nights, total_guests, room_type_changed, stay_category.
- **Encoding:** Categorical variables (e.g., hotel, deposit_type, market_segment) were transformed using **One-Hot Encoding** with the first category dropped to prevent multicollinearity.
- **Scaling:** Numerical features were scaled using **StandardScaler** (Z-score normalization) to ensure that features like adr and lead_time contribute equally to distance-based models (Logistic Regression and K-Means).

B. Handling Class Imbalance

- **Stratification:** Given that cancellations typically represent a minority class, we employed **stratified sampling** during the train/test split. This ensures that the proportion of canceled vs. non-canceled bookings remains identical in both the training (80%) and testing (20%) sets, preventing the models from becoming biased toward the majority class.
- **class_weight='balanced'**

logistic regression: This adjusts the log-loss penalty function. Misclassifying a cancellation will penalise the model roughly (2.6times) harder than misclassifying a kept booking, pulling the decision boundary into an mathematically optimized position.

random forest: This calculates the imbalance weights dynamically within each bootstrap sample (tree level) rather than globally, making the splits more sensitive to the minority cancellation class.

K-Means clustering is an **unsupervised learning algorithm**. Its primary goal is to find natural groupings or patterns within the data, not to predict a target variable. When we applied K-Means, we specifically isolated only the **canceled bookings**. By doing this, we were already focusing on a single 'class' (canceled bookings). Therefore, the concept of class imbalance in the context of the

original is_cancelled target doesn't directly apply, as we were not trying to predict cancellation, but rather to understand the characteristics of already cancelled bookings.

C. Validation Strategy

- **Split Ratio:** An 80/20 train/test split was chosen to provide a substantial training set while maintaining a robust test set for performance evaluation.
- **Random State:** A fixed seed (random_state=42) was used across all operations to ensure consistency in results across different runs.
- The models were trained on the training set, and their performance (accuracy, precision, recall, f1-score) was then evaluated on the unseen testing set.

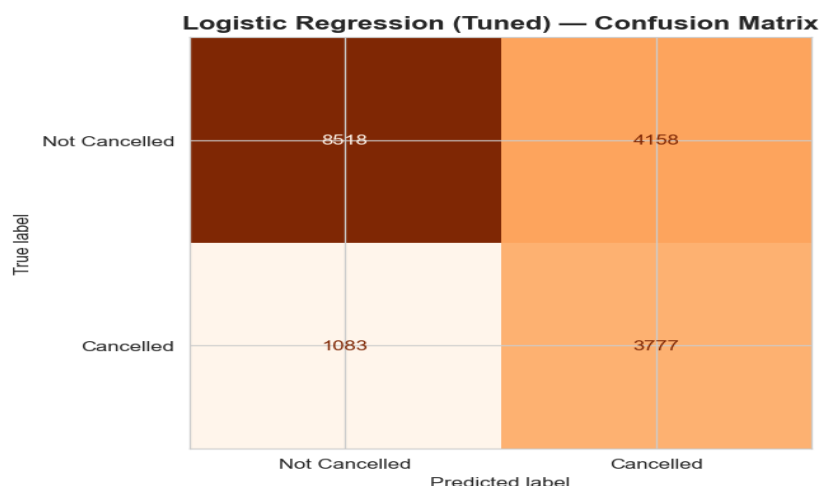
D. Final Hyperparameter Choices

- Logistic Regression:
solver='liblinear', max_iter=500, class_weight='balanced', best params C=0.1, penalty='l1'
- RandomForest:
n_estimators=200, max_depth=20, min_samples_split=5, class_weight='balanced'
- KMeans:
n_clusters=4, n_init = 10

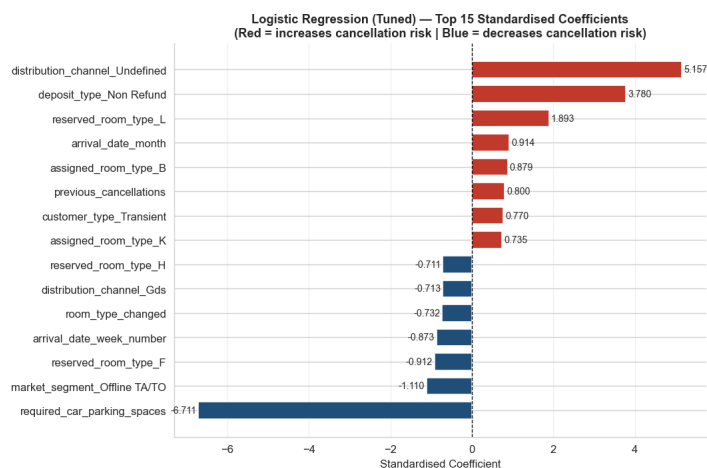
4.2 Results

A Supervised Methods Performance

Logistic Regression Performance: Logistic Regression achieves $F1 = 0.5904$ and $ROC-AUC = 0.8153$. While these are lower than RF overall, LR delivers the highest Recall of the two supervised models, meaning it catches more actual cancellations at the cost of more false positives. This trade-off reflects LR's tendency toward higher sensitivity when class_weight='balanced' is applied with a linear boundary. In the confusion matrix, LR produces 3,777 true positives, it misses fewer cancellations but raises significantly more false alarms.



Accuracy	0.7011
Precision	0.4760
Recall	0.7772
F1 Score	0.5904
ROC-AUC	0.8153



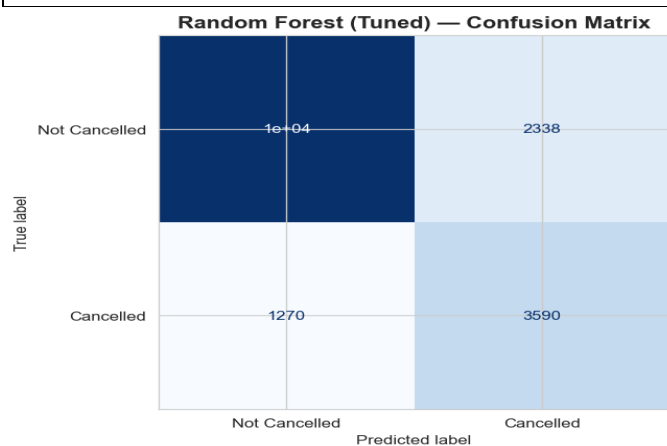
The standardised coefficient plot provides the directional insight that RF feature importance cannot. The most important findings are:

- **deposit_type_Non Refund (positive, large):** The strongest positive predictor of cancellation, consistent with Stage 1's finding of a ~95% cancellation rate for Non Refund bookings. Despite ranking lower in RF's Gini importance, LR correctly identifies it as the highest-impact policy lever.
- **lead_time (positive):** Increasing lead time significantly raises cancellation probability, each standard deviation increase in lead_time raises the log-odds of cancellation. Confirms the Stage 1 EDA analysis.
- **total_of_special_requests (negative):** The strongest negative predictor. Guests who specify dietary requirements, room preferences, or accessibility needs are substantially less likely to cancel — they have invested effort in the booking.
- **is_direct_booking (negative):** Direct bookings (no agent) are less likely to cancel than agent-mediated ones, confirming the rationale for the engineered binary flag.
- **deposit_type_Refundable (negative):** Counter-intuitively, refundable bookings are less likely to cancel than No Deposit bookings, suggesting that guests who explicitly choose a refundable option are more committed to their plans.

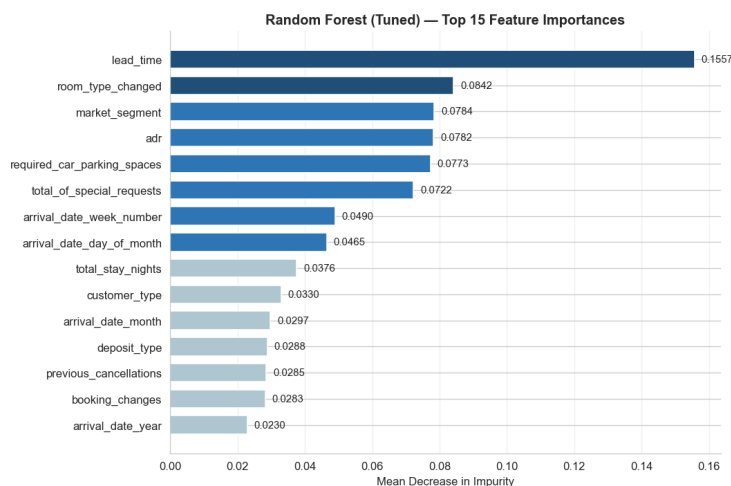
Random Forest Performance:

Accuracy	0.7943
Precision	0.6056

Recall	0.7387
F1 Score	0.6656
ROC-AUC	0.8643

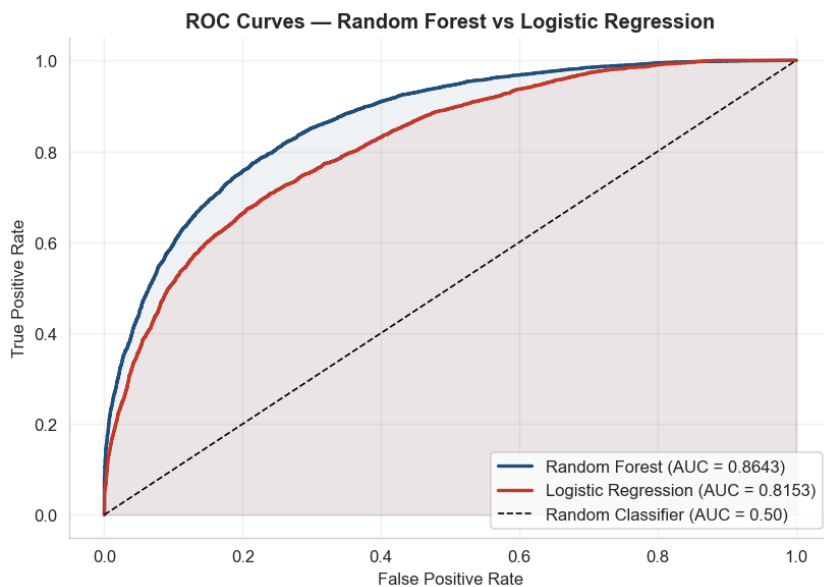


The tuned Random Forest achieves the best overall performance with F1 = 0.6656 and ROC-AUC = 0.8643, substantially above the random baseline of 0.50. The confusion matrix reveals an asymmetric error pattern: 3,590 true positives (cancellations correctly identified) against 1,270 false negatives (missed cancellations). The false negative rate of 26.1% ($1,270 / 4,860$) represents the primary business cost, bookings that will cancel but are not flagged, leaving the hotel exposed. Precision of 0.6056 means that when the model raises a cancellation flag, it is correct 60.6% of the time, a meaningful signal for revenue managers to act on.



The feature importance chart confirms the Stage 1 EDA findings with quantitative precision. `lead_time` is the dominant predictor. Notably, `total_of_special_requests` and `required_car_parking_spaces` both proxies for guest commitment at booking time, rank higher than several categorical features, suggesting that what guests do at booking time is as informative as who they are. The engineered feature `room_type_changed` contributes meaningfully, validating its inclusion in the feature engineering step. `deposit_type` ranks lower than expected due to the known Gini importance bias toward high-cardinality continuous features, its true impact is better captured by the Logistic Regression coefficients.

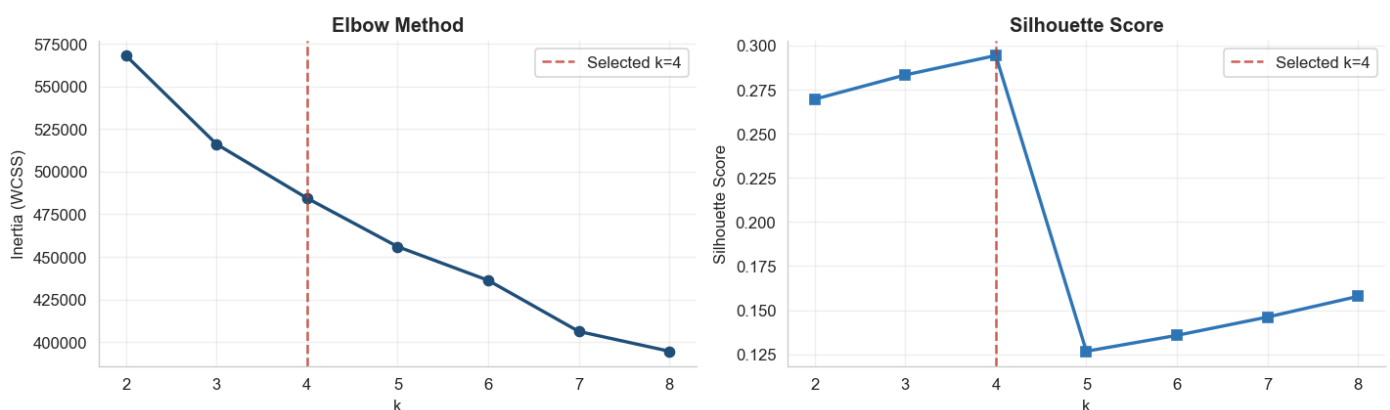
ROC Curve Comparison:



The ROC curve comparison visualises the trade-off between True Positive Rate (sensitivity) and False Positive Rate across all possible classification thresholds. Random Forest (AUC = 0.8643) dominates Logistic Regression (AUC = 0.8153) across nearly the entire curve, meaning RF more reliably ranks cancelled bookings above non-cancelled ones regardless of the threshold chosen. Both models substantially outperform the random baseline (AUC = 0.50). The gap between the two curves is largest at low false positive rates, at FPR = 0.10, RF achieves approximately 0.60 TPR while LR achieves approximately 0.50 TPR, indicating that RF is particularly superior when precision is prioritised (i.e., when the hotel only wants to flag the most confident cancellations).

B. Unsupervised Methods Key Findings

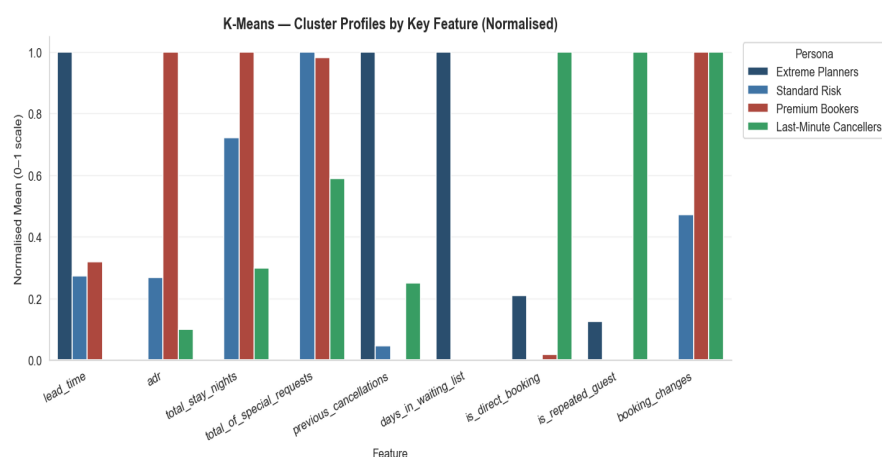
K-Means — Cluster Count Selection



The Elbow method shows a clear inflection point at $k=4$ where the rate of inertia reduction decelerates significantly. The Silhouette score peaks at $k=4$, confirming that $k=4$ produces the most compact and well-separated clusters. The tuning comparison of initialisation methods showed k-means++ consistently outperforms random initialisation, and $n_init=10$ produces a stable solution with no improvement from $n_init=20$ or 50.

Final configuration: k=4, init='k-means++', n_init=10 | Silhouette Score = 0.2943 | Inertia = 484,444

Persona	Size	Lead Time (days)	ADR (£)	Special Requests	Prior Cancels
Extreme Planners (C0)	1,180 (4.9%)	225.9	94.3	0.02	0.46
Standard Risk (C1)	17,607 (72.5%)	105.3	113.5	0.58	0.04
Premium Bookers (C2)	3,058 (12.6%)	113.0	165.7	0.57	0.02
Last-Minute Cancellers (C3)	2,452 (10.1%)	60.2	101.4	0.35	0.13



The four clusters correspond to four distinct cancellation risk personas:

- Cluster 0 — Extreme Planners (4.9%, n=1,180):** Defined by the longest lead times (mean = 225.9 days), the highest prior cancellation rate (0.46 per guest), and near-zero special requests (0.02). These guests book far in advance, have a history of cancelling, and show little engagement with the hotel at booking time. They represent the highest individual risk profile despite being a small segment. Recommended intervention: mandatory non-refundable deposit at booking for lead times exceeding 150 days.
- Cluster 1 — Standard Risk (72.5%, n=17,607):** The dominant segment with moderate lead times (105.3 days), moderate ADR (£113.50), and the highest special request rate (0.58). These are typical Online TA bookings with standard cancellation behaviour. They cancel frequently simply because of their volume. Recommended intervention: targeted email retention campaigns 30 days before arrival for bookings in this segment with lead times above the median.
- Cluster 2 — Premium Bookers (12.6%, n=3,058):** Characterised by the highest ADR (£165.71) and moderate lead time (113 days). Their cancellation represents the largest individual revenue loss per booking. Despite reasonable special request rates (0.57), they cancel — possibly due to price sensitivity or better offers found closer to arrival. Recommended intervention: loyalty programme incentives and price-match guarantees to reduce last-minute comparison shopping.

- **Cluster 3 — Last-Minute Cancellers (10.1%, n=2,452):** The lowest lead times (60.2 days) and the highest rate of direct bookings (0.57), meaning these guests bypass agents and book directly, then cancel shortly before arrival. Their prior cancellation rate (0.13) is moderate. This is the most operationally disruptive group — short lead times leave little time to re-sell the room. Recommended intervention: stricter cancellation windows for direct bookings, with incentives for early confirmation.

A Silhouette score of 0.2943 is consistent with real-world behavioural clustering, guest behaviour does not form perfectly separated clusters, and some overlap between personas is expected. The clusters are internally coherent enough to derive business-meaningful profiles, even if the boundaries are soft.

4.3 Comparative Analysis :

A Quantitative Comparison (Predictive Accuracy)

Random Forest is the stronger classifier by every metric except Recall. Its ROC-AUC of 0.8643 versus LR's 0.8153 reflects a 6.1% improvement in discriminative ability, meaningful in a high-volume operational context where small probability shifts translate to thousands of booking decisions per year. RF's higher Precision (0.6056 vs 0.4760) is particularly valuable when intervention resources are limited: hotel revenue managers cannot review every flagged booking, so a higher-precision model reduces wasted effort on false alarms.

LR's higher Recall (0.7772 vs 0.73870) means it catches more actual cancellations, potentially valuable in scenarios where the cost of a missed cancellation (lost revenue, stranded room capacity) exceeds the cost of a false alarm (unnecessary outreach to a reliable guest). The choice between models in production depends on this cost asymmetry, which varies by hotel type and season.

B Insight-based Comparison (Predictive vs. Descriptive)

- RF and LR predict whether a booking will cancel at the individual booking level; K-Means reveals the structure within the pool of cancellations at the segment level. This distinction matters for strategy: prediction models enable reactive risk mitigation (flag high-risk bookings as they arrive), while clustering enables proactive policy design (design different deposit and cancellation policies for each persona).
- K-Means cluster profiles provide a post-hoc validation of the supervised models' feature importance findings. The feature importance chart shows lead_time and adr as dominant — and the K-Means personas are themselves principally differentiated by lead_time and adr (Extreme Planners vs Premium Bookers vs Last-Minute Cancellers). This cross-model consistency strengthens the overall analytical conclusion.

4.4 Discussion:

A. Performance and Methodological Best Fit

- For the classification task, Random Forest is the best-performing model. Its ROC-AUC of 0.8643, demonstrates strong and reliable discriminative ability. Interpretability is sacrificed, but the feature importance output partially compensates by providing a prioritised list of predictive factors.
- Logistic Regression remains essential despite lower accuracy. Its coefficient output directly links model predictions to policy decisions, a hotel revenue manager can act on 'Non Refund bookings with long lead times are the highest cancellation risk' in a way that a feature importance bar chart alone does not enable.

- K-Means adds a layer of strategic value that neither supervised model provides — it transforms the abstract prediction of cancellation into four named personas with distinct recommended interventions, directly operationalising the research question's goal of informing targeted revenue management strategies.

B. Trade-offs: Interpretability vs. Performance

There is a notable trade-off between the two supervised methods:

- **Logistic Regression** offers high **interpretability**. For management, it is easy to explain how a specific increase in price directly impacts the probability of cancellation. However, its lower performance suggests it may oversimplify the data.
- **Random Forest** provides superior **predictive power** but acts as a 'black box'. While it yields better results for operational decisions (like overbooking strategies), it is harder to communicate the specific 'why' behind an individual prediction.

C. Relation to the Research Problem

By integrating these findings, the hotel can move from reactive to proactive management. The predictive models address the **revenue risk** by allowing the hotel to adjust inventory in real-time. Simultaneously, the K-Means analysis transitions the research into **behavioural strategy**, allowing the hotel to categorize cancellations into actionable personas rather than viewing all cancellations as equal risks.

5. Conclusions and Limitations

5.1 Key Findings

This analysis has successfully addressed the research problem: "**How can the hotel accurately predict the probability of cancellation, and what distinct 'risk personas' exist among guests that contribute to revenue instability?**"

Stage 1 findings:

- **Lead time is the strongest cancellation signal:** Stage 1 EDA revealed a monotonic relationship between booking lead time and cancellation probability, rising from 8.45% for same-week bookings to 38.42% for bookings made more than 150 days in advance. This single finding has direct implications for tiered deposit policies.
- **Deposit type is counter-intuitive but highly predictive:** Non Refund bookings show a cancellation rate approaching 95%, far exceeding No Deposit bookings. This finding challenges the assumption that financial penalties deter cancellations and motivates the exploration of alternative deterrence mechanisms.
- **City Hotels face a disproportionate cancellation burden:** Both in volume and rate, City Hotels cancel more than Resort Hotels, likely reflecting the higher proportion of Online TA and business traveller bookings in the former.
- **Data quality issues required careful handling:** 31,328 duplicate rows (26.1% of raw data), leakage variables (reservation_status, reservation_status_date), and high-cardinality features (country: 178 values) were identified and resolved before modelling.

Stage 2 Findings

- **Random Forest is the best classifier:** ROC-AUC = 0.8643, F1 = 0.6656. After hyperparameter tuning (max_depth=20, n_estimators=200), the model correctly identifies 73.9% of actual cancellations with 60.6% precision, providing a deployable risk scoring tool.
- **Logistic Regression confirms feature directions:** Non Refund deposit type and lead_time are the strongest positive predictors of cancellation; total_of_special_requests and is_direct_booking are the strongest negative predictors. These directional findings translate directly into revenue management policy.
- **Four cancellation personas were identified:** Extreme Planners (very high lead time, repeat cancellers), Standard Risk (the dominant group, typical Online TA behaviour), Premium Bookers (highest ADR, revenue-critical), and Last-Minute Cancellers (direct bookings with short notice). Each persona warrants a distinct intervention strategy.
- **Feature importance is cross-validated across models:** lead_time and adr rank highest in RF importance and appear as the primary dimensions of differentiation in K-Means clusters, providing cross-method validation of these features as the key drivers of cancellation risk.

5.2 Limitations

Dataset Limitations

- **Temporal scope:** The dataset covers July 2015 to August 2017. Consumer booking behaviour has changed substantially since then. Models trained on this data may not generalise to the current hospitality landscape without retraining.
- **Geographic scope:** Both hotels are in Portugal. The findings, particularly around deposit type behaviour and booking channel preferences, may not generalise to hotels in other markets with different cancellation cultures or regulatory environments.
- **Missing external context:** The dataset contains no information about competing hotel prices, local events, or macroeconomic conditions at the time of booking, all of which influence cancellation decisions. The model therefore captures booking characteristics but not the environmental conditions that motivate cancellation.

Methodological Limitations

- **K-Means cluster interpretability:** The Silhouette score of 0.2943 indicates moderate rather than strong cluster separation, behavioural segments in real-world data naturally overlap. Cluster membership should be treated as a probabilistic assignment rather than a definitive categorisation, and persona names are interpretive rather than ground-truth labels.
- **Gini importance bias in RF:** Mean Decrease in Impurity systematically overestimates the importance of high-cardinality continuous features (lead_time, adr) relative to binary or low-cardinality categorical features (deposit_type, is_direct_booking). A permutation importance analysis would provide a less biased ranking but was beyond the scope of this project.
- **Linear assumption of Logistic Regression:** LR assumes a linear relationship between features and log-odds of cancellation. The actual relationship, particularly for lead_time, which shows a non-linear step change between booking windows, is unlikely to be perfectly linear. This partly explains LR's lower accuracy relative to RF.

6. Contribution Statements

Both group members contributed equally to all sections of the report and code and worked collaboratively throughout the project. Each member participated in the topic selection, data preparation, exploratory data analysis, modelling, experimental evaluation, result interpretation, report writing, and code development. All sections were jointly discussed, implemented, reviewed, and refined together to ensure consistency and equal contribution across the entire assignment.

Appendix

AI Acknowledgement: ChatGPT and Gemini were used to assist with the report structure, paraphrasing, and grammar correction.

ChatGPT and Gemini were also used to help generate parts of Python code and to troubleshoot errors.

All AI-generated content was carefully reviewed, edited, and validated before inclusion.
